1. **AI/ML Applications in RAN:**

Give one example of how AI/ML can improve radio resource management and one example of how it can enhance network reliability in Open RAN.

AI/ML can improve radio resource management by dynamically allocating bandwidth based on user demand and network traffic. It can improve network reliability by detecting network faults early and automatically adjusting resources to prevent service disruptions.

2. **SON and Predictive Maintenance:**

Describe how self-optimization and self-healing work in Self-Organizing Networks (SON). How does predictive maintenance improve Quality of Service (QoS)?

Self optimization automatically adjusts network settings to improve performance, while self healing detects and fixes network issues with minimal human intervention. Predictive maintenance improves QoS by identifying potential equipment failures before they cause downtime or service degradation.

3. **Hands-On Lab Application:**

In the Python-based resource allocation simulation lab, what does normalizing user demand achieve? How does this relate to real-world resource scheduling in Open RAN

Normalizing user demand scales values to a common range so resources can be allocated fairly among users. In Open RAN, this is similar to resource scheduling where network resources are distributed efficiently based on user requirements and network conditions.

4. **Traditional RAN vs. Open RAN:**

Explain two key differences between traditional RAN and Open RAN architectures. How does Open RAN promote vendor interoperability?

Traditional RAN uses proprietary hardware and software from a single vendor, while Open RAN uses open interfaces that allow components from different vendors to work together. Open RAN promotes vendor interoperability by using standardized interfaces and open architectures.

5. RAN Intelligent Controller (RIC):

Distinguish between the roles of the Near-RT RIC and Non-RT RIC. Provide an example function for each.

The Near RT RIC makes real time decisions such as traffic steering and resource optimization, while the Non RT RIC handles long term tasks like AI model training and policy management. For example, the Near RT RIC can optimize bandwidth allocation, while the Non RT RIC can analyze historical data to improve future network performance.